

TIC 453692345

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.814639 d
Epoch (t0) 1570.6181 BTJD
Duration 0.0580 d
Depth 9.708 % (97,077 ppm)
SNR 397.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 59 (gate ≥ 3)
Robust depth ratio 1.10 (gate ≥ 0.50)
Robust depth sigma 58.8 (gate ≥ 7.0)
Odd/even z-score 1.85 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 25.9 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 25.75 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 453692346
0.29 px away, dTmag +1.69 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

